

Serhii Revva for Fruitful Code, Dribbble

Interactive Development 300: Semester 2

ADVANCED SOFTWARE DEVELOPMENT 2

THEME 4:

Theory: Industry Perspectives
Practical: AI-Driven Application

FINAL PROJECT:

An AI-Driven Application

28 October 2026, 14:00, Github Repository & Documentation

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INTRODUCTION

"AI will change the nature of creativity, so it's important that we develop new ways to interact with these machines." – Yves Behar

We're living through a pivotal moment in technological history; one that's rapidly reshaping the way we work, create, and interact. At the heart of this transformation is artificial intelligence (AI).

As developers, this shift is directly tied to our discipline. Whether or not you plan to specialise in AI, understanding its foundations and implications is quickly becoming essential. More and more of what we build, relies on or is enhanced by AI-driven tools.

You've likely already experienced the benefits of AI through tools that assist with code, automate workflows, or help streamline design. The industry is moving fast, and with massive investments in AI, developers now have access to a growing ecosystem of frameworks, APIs, and cloud-based services.

For this theme, you'll have the opportunity to explore how AI can be integrated into real-world development projects. Alongside this, we'll place a strong focus on professional growth. As you prepare to enter the industry, learning from those who've already navigated its challenges, and helping you make more informed, strategic decisions about your own path and work. What you will learn:

- Exploration of AI
- Implementation of various AI services
- Freelance project management
- Engage & reflect on discussion with industry professionals

PROJECT CONTEXT

In recent years, AI has emerged as a driving force behind innovation across nearly every industry. By leveraging massive datasets and advanced algorithms, AI systems can identify patterns, generate insights, and make decisions at speeds and scales far beyond human capabilities. From accelerating breakthroughs in cancer research to supporting solutions in climate change, AI is increasingly shaping the way we solve complex global challenges.

For this project, you'll research a real-world problem in one of the following domains:

- Social Impact
- Accessibility
- Education
- Environment,
- or Healthcare.

You'll then conceptualise, design, and develop an AI-powered application that seeks to create meaningful change in your chosen area.

This project serves as the final addition to your portfolio and should showcase your ability to apply development skills in a way that is both technically sound and socially relevant. You're free to use any programming language, framework, or environment, as long as your solution incorporates at least one AI service as a core feature, not just a "nice-to-have." Your goal is to build something that could realistically be deployed and have a positive, measurable impact.

LEARNING OUTCOMES

On completion of this semester's *practical* deliverables, you should be able to:

- Understand and explore the importance of Artificial Intelligence in the development industry
- Identify a real-world problem and implement an AI project to aid as a solution to the problem
- Manage a real-world project with appropriate software design practices and understand the process of implementing an independent project
- Develop a real-world project using already available AI services by leading companies, such as Amazon, IBM, OpenAI and Google
- Improve on the user experience of a development project based on conducted user tests and feedback
- Understand the fundamentals of the Python programming language and how it can be used in conjunction with AI platforms
- Understand the implementation of machine learning algorithms

On completion of this semester's *theory* deliverables, you should be able to:

- Engage with industry professionals and the different roles in the development industry
- Gain information on the positions, tasks, workflows and companies associated with the development field
- Identify and reflect on their perceptions and expectations regarding industry and their future career paths

SEMESTER PHASES & TASK SUMMARY

During your *practical* project, you will go through the following phases & assignments:

1 PROJECT PITCH

Formative - Project Pitch | **Week 10** | **PDF**

To start with this theme project, you will first need to plan and conceptualise what the theme of your app will be. You will need to consider the real-world context, the technology that will be used to build the app and the end-users, to help you plan the structure of data implementation as well as the overall design of the app. This will need to be compiled into research summaries, diagrams, moodboards and wireframes, which your lecturer will provide you feedback on.

2 IMPLEMENTATION PROGRESS

Formative - Progress Milestone | **Week 12-14** | **Live**

For the project, you will implement your AI-driven application utilising any programming language or framework of your choice, as well as at least one AI service that forms a part of your main feature. This will be guided by your project pitch and planning. Make sure that the tools you choose to use is the best suited for the type of project you build and for the purpose of the app.

During the implementation of your project, you will have a review session, in which you will be receive feedback on your progress and code quality. These sessions will be conducted in class or in contact sessions.

3 DEMONSTRATION OF FINAL APPLICATION

Summative - AI-Driven Application | **Week 15** | **Git & Live**

As the main project, you will implement your AI-Driven Application. The brief will need to be analysed to ensure that all technical requirements are being met, and at the same time, you will need to research and make various discussions based on the real-world context your app will address.

At the end of the semester, you will need to present the entire application with a breakdown of the implementation approach and a live demonstration of the application. After your demonstration, you will be asked to explain specific features and code. Your up-to-date repository will be your submission with appropriate documentation.

For *theory*, you will have the opportunity to chat to actual devs in industry:

INSIGHTS & REFLECTION VIDEO

Theory - Reflection Video | **Week 15** | **mp4 / mov**

To conclude theory, the course is designed to be dynamic, featuring industry speakers from various avenues within the field of DV who will share their knowledge and experiences. The aim is to inspire and guide you as you make decisions regarding your future careers within the field.

You will be required to create a reflection video, identifying the key findings of our industry speakers and reflecting on your studies at Open Window.

PROJECT PITCH

PRACTICAL

There are very few limitations with this project, but that means the responsibility will be on you for the decisions you make. These decisions will be guided through your research, planning and the class activities you complete each week.

In this first phase of the project, you will be spending time researching various problems and identifying what application you would like to build to solve this problem. This will include planning the type of AI technologies and the frameworks, as well as any technical planning, such as the user-flow and the scope of the project.

As part of the user interface implementation, you will also be required to create a brand identity and aesthetic for the frontend of you app. This includes logos, copy, appropriate imagery and icons, along with standard UI styling. As part of you research and planning, you will be required to create moodboards, wireframes and style guides.

All of this planning & research needs to be compiled into a project proposal.

Proposed Pitch Breakdown

Your pitch document should be concise, visual, persuasive and will be submitted as a PDF. It should reflect intentionality and a clear understanding of what makes your app different and achievable.

Please remember to include appropriate referencing in the document.

Here is a breakdown of the recommended sections:

A) PROJECT TITLE

- Identify the name of the app
- A one-line description (like a product tagline)
- App's Logo

B) CONCEPTUALISATION & PROBLEM STATEMENT

- Overview of what the app will be
- The core problem or need your app is trying to address (explain in which domain it will function). This will require additional research.
- Who is this app for? Why would they care?
- Then, identify the type of application you can create to solve this problem and why this type of app

C) CORE FEATURES

- Define the scope of the application: What are the minimum viable features you plan to implement?
- Define SMART objectives
- Prioritise features as MVP (Must-Have), Nice-to-Have, Future Considerations
- What will the different screens/pages of the application be?
- Define the different user roles (e.g., Admin, Standard User, Read-Only User)
- Userflow-diagrams or user stories to plan various use cases of application, while considering different user roles

D) TECHNICAL PLANNING

- Identify the type of AI technology that you would need to implement your application.
- AI Service: Identify which AI service you will use in your application and how it will form part of your app's main feature
- Technologies & Tools: Briefly explain which technologies you will use to build the application and why.
- Mention any technical challenges you foresee and how you'll approach them
- If your project requires a backend, you will be required to plan the structure of your data either by identifying the core database schemas or creating an Entity Relationship Diagram (ERD).

PROJECT PITCH

PRACTICAL

E) WIREFRAMES & UI/UX CONSIDERATIONS

- Find inspiration for the front-end styling and identity of the website that will best represent the data that will be implemented. The inspiration should be in the form of moodboards and can include any inspiration (we recommend looking at UI designs specifically).
- Include rough sketches or wireframes for your main screens (remember that this is a mobile app)
- Justify any frontend frameworks you will use, or identify general styling decision for the project
- Indicate any accessibility or usability concerns

F) PROJECT ROADMAP

- Estimated what the roadmap to complete the application might look like, considering the different phases and deadlines of the semester
- Remember, it is important to make continuous progress on your project and to follow best practices when it comes to version controlling, so keep this in mind as your plan out the different phases of your implementation.

G) CONCLUSION

- Final remarks on why this app is suitable

**** This assignment acts as a qualifier for your progress milestone assessment. If you do not submit this document, you will still receive feedback during the progress review, but will receive 0%.*

ASSESSMENT CRITERIA

THEORY - PROJECT PITCH

Brand Conceptualisation: 20%

- Does the student demonstrate a capacity to interpret different inspiration and limitations to conceptualise an application idea and it's required features clearly?

Concept Planning 40%

- What is the impression of the student's app planning in terms of the concept idea, app name, moodboard, colour palette and basic wireframes?
- Has the student appropriately considered the userflow and experience of the app with appropriate diagrams?

Technical Planning 40%

- Has the student appropriately considered the structure and relationships between data with appropriate diagrams?
- Has the student clearly identified the requirements and functionality of the app, while considering the theme of their app?

Monday, 21 September 2026 at 23:59 - W10 PDF

PROGRESS MILESTONES

PRACTICAL

Throughout the theme, we'll be conducting progress feedback and code reviews to help track your development and ensure you're on the right path to complete the project. Each of you will have individual check-ins for grading and feedback.

You will randomly be selected to showcase your progress and have a code review in one of these weeks. Therefore, you will need to ensure that you properly plan and manage the sprints of your projects to make continues progress and to be ready for your stand-up meeting in either one of these weeks.

Both the milestones and code reviews are designed not just to assess your work but also to guide your learning. If you do not continuously work on this project, you will not receive marks.

A) PROGRESS MILESTONE

This milestone will involve a one-on-one check-in where you'll present your implementation, discuss your progress, and demonstrate key features or functionalities you've implemented. Your lecturer will provide feedback and assign a mark based on your understanding and development so far.

B) CODE REVIEW

During this session, we'll dive into your actual code. You'll either walk your lecturer through your coding decisions, structure, and problem-solving approaches, or your lecturer will assess your code via your GitHub commits. The focus will be on code quality, maintainability, and how well you've applied the concepts we've covered in class.

C) USABILITY TESTING & PEER REVIEWS

In Week 15, your fellow peers will test your MVP and give you feedback on the product experience and usability, based on a review form that you will need to create. These suggestions and feedback will need to be analysed and utilised for your final product. This is a great opportunity to get a different perspective on your project, and also to refine your project for your final submission and portfolio.

ASSESSMENT CRITERIA

FORMATIVE - PROGRESS MILESTONE

Functionality Progress: 25%
- Are all implemented features functional and meet requirements? Is it clear that the student understands the functionality?

Sprint Planning & Agile Execution: 25%
- Has the student met the requirements stipulated in the scope planning? Is the student on schedule or ahead of schedule? Does the student know what needs to be done next?

Code Quality & Organisation: 20%
- How clean, modular, and easy to follow is the student's code? Is the students code well-organised? Has the student adhere to industry best practices in terms of naming conventions, DRY principles etc? Has the student done/written appropriate testing of the code?

Code Understanding: 15%
- Is it clear that the student understands their implemented code?

Version Control: 15%
- Has the student effectively utilized git for version control? Was version control implemented with appropriate commit messages, clear branching strategies, and regular updates?

Week 12-14 | Live Demo | GitHub

FINAL AI-DRIVEN APPLICATION

PRACTICAL

Build an AI-Driven Application that addresses a specific problem, in any tech environment, and using at least one AI service of your choice.

This deliverable is the end goal of this theme.

Project Specifications

A) Project Limitations, Planning & Research

For this project, you will be required to do appropriate research on any problem that you would like to address with your application. This will require research on the potential users, what their needs are, how your application can aid in those needs, as well as how the user will use the application in terms of functionality and frontend. You will also need to research any appropriate technologies and AI services that will be suited to build your application. This technology research includes required functionality for your application.

There are very few limitations with this project, but that means the responsibility will be on you for the decisions you make. These decisions will be guided through your research, planning and the assignments you receive each week:

- **Any Application Type**
  The application can be a web app, a mobile app, or any other application that includes an interface, i.e. not just a console/terminal app. Consider the user that you are building your application for, and what type of application would make the most sense for the impact your application tries to achieve.

- **Any Appropriate Languages/Frameworks**
  You can use any development technologies that you have learnt in Interactive Development (DV100-DV300).
  You are also allowed to use other technologies that you have not worked with yet, if approved by your lecturer. Only important requirement is that the development environment is appropriate for the application you want to build.

  If your project requires some sort of backend, you will be required to plan your database appropriately based on the type of application you will be building. Again, you can utilise any backend technologies.

- **AI Technologies**
  During this course we will explore and look into the development of machine learning and deep learning, as well as various ways to deploy machine learning models to be used by our applications. ***However, you will not be required to build your own AI model (unless your project idea requires it).*** Instead you will be introduced to multiple existing cognitive services through the term that you can use, such as OpenAI, IBM Watson, Microsoft Azure's Cognitive Services, AWS AI Services, Google Cloud AI services, etc. For your final project, you are required to use **at least one AI service** (which does not have to be covered in class) as one of your project's **main features**.

- **Screens/Pages Requirement**
  You will be required to plan and implement a **minimum of five (5) screens/pages** that your application will need, each with the required functionality and sections. Consider any onboarding, terms of usage and authentication screens your application might need for user experience.

B) The Theme, Brand Identity & Scope

You are expected to plan and build your AI-driven application based on the problem it will address. This allows you to drive the overall thematic direction of your project, and you will ultimately be responsible for the foundational requirements of your project to make it usable in the real-world. This includes product naming, branding, UI/UX design, and any additional features that align with the problem your app will address. Your application will need to have a designed app icon or logo that follows the guidelines stipulated by the respective platforms that you are building for.

The Project Pitch assignment will serve as a guide to structure your development approach, ensuring that the core requirements are met while allowing for creative freedom. You will receive feedback and suggestions from your lecturer and peers, and you should be prepared to adapt to these evolving requirements while managing your scope effectively.

FINAL AI-DRIVEN APPLICATION

PRACTICAL

C) Project Management, Progress Milestones & Peer Reviews

Make sure to plan what the minimum viable product (MVP) of your project needs to be. Schedule your sprint meeting according to this MVP product and leave extra time to refine the functionality and add "nice-to-haves". We will approach this project using an agile methodology throughout the term and for your progress marks, which we will handle as sprint stand-up meetings, requiring you to explain your progress on the project each week.

As already mentioned, you will receive feedback from your lecturer on your project idea, as well as from your peers. Make sure to document any feedback received and how it was incorporated into the project. Additionally, you will participate in a progress milestone and usability testing session where you'll present the current development status of the project and receive feedback. This will include a code review to ensure code quality and best practices are being followed.

D) Project Deployment

Since the project is real-world focused, we would like to encourage you to deploy and publish the project you create so that people can access and use your amazing app (Open Window will assist with any appropriate services). This will be assessed as a bonus mark.

E) GitHub Repository & Version Control

For each project, you are required to utilise Git and GitHub in order to implement version control and collaboration into your projects. You must add your lecturer as a collaborator at the start of all project repos. Regular pushes are expected. Make sure that your repository contains the latest commits in the **main** branch.

Your assessors will assess the frequency, consistency and quality of your commits over the term. Your GitHub repository will be the final hand in format for your semester projects.

F) ReadMe Documentation & Demonstration Video

Proper **documentation** is essential for both usability and maintainability of the system. This forms part of your portfolio requirements for the end of the year, and therefore, you are required to create a comprehensive **README** file, which should include:

- A **project overview** explaining the purpose and features of the system.
- **Installation instructions** for setting up the backend, database, and frontend on different environments (for your lecturer to test the project).
- **Features/Usage instructions** detailing how to navigate and use the application, including descriptions of key features.
- Any **mockups or diagrams** that can form part of the documentation and give better context to the implementation.
- Information on **technologies used** and contributions from each team member.
- An **acknowledgement** section to reference any resources used for this project.

Additionally, you must include a **demonstration video** in the documentation to showcasing the system's functionality. This 5-8 minute video should be a walkthrough of the key features and demonstrate user interactions (e.g., login, dashboard navigation, CRUD operations).

NB! When making a screen recording, please ensure that you only make a screen recording of your app and/or code. Crop out background elements such as your desktop or iOS / Window menu bars. You want your video to display professionally with no background sound or distracting windows.

G) Live Project Demonstration

At the end of the semester, you will be required to present the final application in class. This will include any relevant information in terms of the project approach, a demonstration of implemented features, as well as an opportunity for you to explain any of the implemented code..

** have your video demonstration on stand-by for any technical issues*

FINAL AI-DRIVEN APPLICATION

PRACTICAL

ASSESSMENT CRITERIA

SUMMATIVE - FINAL AI-DRIVEN APPLICATION

Project Implementation & Completeness: 65%

- **Technology Requirements:** Are the chosen technologies and frameworks appropriate for the application? How well has the student implemented the fundamentals of implementing a project with the selected technologies?
- **Functionality Refinement:** To which level has the features of the application been refined? Are all functionality/screen requirements being met, with minimal bugs?
- **AI Implementation:** How effectively is the AI integration implemented as a core feature? Is the AI service selected the best solution for the application requirements? Does the project include at least one AI service?
- **CRUD Functionality:** How efficient is the displaying of data? How comprehensive and efficient is adding, manipulation and deleting of data?
- **Security & Authentication:** How comprehensive has the student implemented security practices or authentication functionality in the application?
- **Project Management & Feedback:** Has the student demonstrated consistent progress toward their MVP across sprints? How effectively has the student responded to and incorporated peer and lecturer feedback?
- **Deployment (Bonus):** How ready is the application for deployment? Has the student deployed the application to an appropriate platform?

Application Concept: 20%

- **Real-World Context:** Does the project showcase & address a clearly defined problem within the chosen realm? How significant and relevant is the problem being addressed?
- **Thematic Accuracy:** How effectively does the address the identified problem? Is there a well-defined target audience for this application incorporated in the theme/concept of the application?
- **Usability:** Is it clear who the application is for, and is the application meeting the outlined feature requirements of their concept? Is the application easy to use and understand, creating a seamless user experience? Is there clear validation and feedback?

Project Administration: 15%

- **Code Quality:** Is the repository well managed, naming conventions specific and directories well organised? Is the repository's code readable and refactored, following standardised practices?
- **Version Control:** Has the student effectively utilised git for version control? Was version control implemented with appropriate commit messages, clear branching strategies, and regular updates?
- **Documentation:** Has the student included an appropriate ReadMe file with all the appropriate information (including a demonstration video)?

Wednesday, 28 October 2026 at 14:00 - W16

GitHub URL

INDUSTRY INSIGHTS & REFLECTION

THEORY

As you prepare to step into the industry next year, the value of learning from those who have already navigated the path ahead becomes increasingly evident. Engaging with experienced professionals offers insights that extend beyond textbooks, providing real-world perspectives that can shape your career choices.

For our theory class, you will engage with industry speakers and reflect on your learning in the form of a video essay.

You will be required to create the following for the assignment:

[A] PRESENTATION DOCUMENT

Create a presentation document detailing your insights from speaker sessions and your own expectations about the industry

As third-year university students, you have acquired a wealth of knowledge and skills. The time has come for you to transition from the academic environment into the professional world. You are required to design a document that communicates all the insights and lessons you learned from the variety of industry talks you attended. This assignment also aims to encourage self-reflection on your academic journey and your expectations, aspirations, and apprehensions about entering the industry relevant to your field of study.

- Make sure the presentation is in landscape orientation. Provide a cover page for the document with your name and student number.
- Identify the major lesson / insight you gained from a **EACH** industry speaker / talk. Summarise the insight into a single paragraph.
- Create new slide to communicate **EACH INSIGHT** and include a pull quote from each talk to accompany the insight gained from that speaker. In essence each slide of the deck should contain a single insight and a pull quote from a speaker.
- Then include at least **TWO** slide to explain what you want to do in industry, what your expectations are, and reflect on your academic journey.
- You need a minimum of 6 insight slides, and 2 reflection slide.
- **THIS IS NOT AN ESSAY**! This should be a carefully consider document with a layout and use of text and imagery. Ensure it is a visual aid to your video response!

[B] VIDEO RESPONSE

Using your presentation document, create a video response explaining the insights of the speakers and your expectations and ideas about the industry

Create a **5-8 minute** video presentation in which you highlight the key insights from the different industry talks, as well as reflects on your experiences and thoughts as you prepare to transition into the industry related to your field of study. You are encouraged to be creative and thoughtful in your presentation.

VIDEO KEY POINTS

- Briefly introduce yourself
- What were your key lessons/insights of each speaker/talk?
- What do you want to do? Share your career aspirations and goals entering industry.

INDUSTRY INSIGHTS & REFLECTION

THEORY

- Discuss your expectations of industry. What are your hopes and potential challenges?
- Reflect on your academic journey and HIGHLIGHT ONE moment that has been particularly meaningful or impactful that you feel has shaped you for industry. This could be a course, project, conversation - anything!
- Conclude your video with any final thoughts

You as an audience member of the industry talks will also be assessed on your contribution to the class discussion.

PLEASE NOTE:
The speaker information will be available on Canvas by 12 September 2026, with links to the online Google Meetings. These sessions will not be recorded.

ASSESSMENT CRITERIA

THEORY 4 - INSIGHTS & REFLECTION VIDEO

Report Document: 25% PDF

- **Spelling and communication:**
  Is the writing and spelling executed at a high standard?
- **Substance, Depth and Effort:**
  Is the content explored in a substantial manner?
- **Reflective Capacity and Insight:**
  Is there a clear engagement and personal interpretation of industry expectations?
- **Text and imagery**:
  Does the report use text and imagery effectively?

Video Response: 65% MP4/MOV

- **Clarity and organisation:**
  Is the video content organised and clear?
- **Reflective Capacity and Insight:**
  Is there a clear engagement and personal interpretation of industry talks & their expectations?
- **Communication:**
  is the communication effective and engaging?
- **Use of media:**
  Does the video use imagery, video and transitions?

Class Participation: 10%

- **Engagement:**
  Has the student participated in at least 3 of the industry talks?

Monday, 26 October 2026 at 23:59 - W15

THEME 3 MARK BREAKDOWN

Phase	Marked Component	Due	Weighting
1	**Project Pitch** (Formative Qualifier)	21 Sep 2025 at 23:59 - W10	0%
2	**Progress Milestone** (Formative)	W12-14	30%
3	**Final AI-Driven Application** (Summative)	28 Oct 2026 at 14:00 - W15	60%
	Industry Insights & Reflection Video (Theory)	26 Oct 2026 at 23:59 - W15	10%
Theme 3 weighting for the semester 2			50%

YEAR MARK BREAKDOWN

Semester 1	30%
Semester 2: Theme 3	15%
Semester 2: Theme 4	15%
Final Portfolio Evaluation	40%

SUBMISSION FORMAT

Use the below checklist as a guideline on submission formats:

📄 **Project Pitch** — WEEK 10

- ✅ PDF format
- ✅ All required information

👩‍💻 **Progress Milestones** — WEEK 12-14

- ✅ Live discussion & demonstration
- ✅ Git Repo up-to-date

🔗 **Final AI-Driven Application** — WEEK 15

🗂️ GitHub Repository Link

- ✅ Commits up to date
- ✅ All files
- ✅ Set to public

🖨️ **ReadMe Documentation**

- ✅ All required content
- ✅ 5-10 minutes demonstration video

📄 **Insights & Reflection Video** — WEEK 15

- ✅ Showcase of key insights from industry speakers
- ✅ Reflection on studies & future
- ✅ PDF slides
- ✅ MP4/MOV video
- ✅ 5-8 minutes video



Rabbit Tech

RESOURCES

RECOMMENDED USEFUL DOCUMENTATION

OpenAI API Documentation
IBM Watson Documentation
Google ML Kit
Amazon ML Services
Microsoft Cognitive Services
LangChain Documentation
Tensorflow Documentation
GitHub Guides
Scrum Guide

VISUAL RESOURCES AND SITES YOU ARE ALLOWED TO USE

Photography: Unsplash, Pexels, StockSnap, Nappy, PikWizard, Burst, KaboomPics, Gratisography

Icons: Lucide, Flaticon, TheNounProject, Icons8, Iconfinder

Inspiration: Pinterest, Dribbble, Behance, Dark,

Mockup Templates: Canva, Mockuuups Studio, Unblast

The above list is safe to use for this project. You may also use photography, icons and mockups from sites that do not require attribution to the original designer and are considered free to use or under the 'Creative Commons 0' agreement. If you can't find specific resources needed, speak to your lecturer for advice and guidance. Feel free to also generate your own images with AI, as long as you reference the model used appropriately.

